

Pipeline Mathematics

Step 1. Time Feature Augmentation

$\mathbf{X} \in \mathbb{R}^{N \times 31}$: raw normalised input, $N = 14,976$ province-week rows, 31 epidemiological, climate, and spatial features. Missing values imputed with column means computed on training years only.

For each row i with week index $w_i \in \{1, \dots, 52\}$:

$$t_i^{\sin} = \sin\left(\frac{2\pi w_i}{52}\right), \quad t_i^{\cos} = \cos\left(\frac{2\pi w_i}{52}\right) \quad (1)$$

$$\tilde{\mathbf{X}} = [\mathbf{X} \mid \mathbf{t}^{\sin} \mid \mathbf{t}^{\cos}] \in \mathbb{R}^{N \times 33} \quad (2)$$

Step 2. Autoencoder Encoding

The encoder half of a trained autoencoder compresses each row:

$$\tilde{\mathbf{X}} \in \mathbb{R}^{N \times 33} \xrightarrow{\text{encoder}} \mathbf{Z} \in \mathbb{R}^{N \times 6} \quad (3)$$

$\mathbf{Z}$ contains the 6-dimensional latent representation of each province-week.

Step 3. State Vector Construction

Raw incidence $\mathbf{y} \in \mathbb{R}^N$ is clipped and log-transformed, then appended as a 7th column:

$$\tilde{y}_i = \log(1 + \max(0, y_i)) \quad (4)$$

$$\mathbf{S} = [\mathbf{Z} \mid \tilde{\mathbf{y}}] \in \mathbb{R}^{N \times 7}, \quad \mathbf{s}_i = [z_1, z_2, z_3, z_4, z_5, z_6, \tilde{y}]_i \quad (5)$$

Step 4. Lag Matrix Construction

$p = \text{n_lags}$ is a hyperparameter discovered by the island search (bounded in $[2, 8]$, seeded at 5).

For every valid time point t within a province, flatten the $p + 1$ consecutive state rows into one wide feature vector:

$$\boldsymbol{\theta}_t = [\mathbf{s}_{t-p} \mid \mathbf{s}_{t-p+1} \mid \dots \mid \mathbf{s}_t] \in \mathbb{R}^{7(p+1)} \quad (6)$$

The prediction target is the full state one week ahead:

$$\mathbf{y}_t = \mathbf{s}_{t+1} \in \mathbb{R}^7 \quad (7)$$

Stacking all valid windows across all provinces:

$$\boldsymbol{\Theta} \in \mathbb{R}^{T \times 7(p+1)}, \quad \mathbf{Y} \in \mathbb{R}^{T \times 7}, \quad T \approx 12,000 \quad (8)$$

For $p = 4$: $\boldsymbol{\Theta} \in \mathbb{R}^{T \times 35}$. The first p rows of each province chunk are dropped (insufficient history).

Lag interpretation. Each of the $p + 1$ blocks in a row of Θ is a different week, so feature z_k appears $p + 1$ times as independent columns. STLSQ fits a separate coefficient to each. After thresholding, only some survive. If z_k at lag $t - 2$ survives but z_k at t and $t - 1$ die, the model has discovered that z_k from two weeks ago drives the dynamics. The lag structure is not imposed. It emerges from thresholding.

Step 5. STLSQ

Objective. Find a sparse Ξ such that:

$$\mathbf{Y} \approx \Theta \Xi, \quad \Theta \in \mathbb{R}^{T \times 35}, \quad \Xi \in \mathbb{R}^{35 \times 7}, \quad \mathbf{Y} \in \mathbb{R}^{T \times 7} \quad (9)$$

OLS closed form (one iteration):

$$\hat{\Xi} = (\Theta^\top \Theta)^{-1} \Theta^\top \mathbf{Y} \quad (10)$$

Term	Shape
$\Theta^\top \Theta$	35×35
$\Theta^\top \mathbf{Y}$	35×7
$\hat{\Xi}$	35×7

Iterative thresholding loop:

1. Compute $\hat{\Xi}$ via OLS.
2. Zero out all entries: $\hat{\xi}_{ij} \leftarrow 0$ if $|\hat{\xi}_{ij}| < \lambda$.
3. Refit OLS on surviving columns only.
4. Repeat until no further entries are eliminated.

The final sparse Ξ is the **discovered equation of motion**. Its surviving entries are the coefficients of the governing equation:

$$\mathbf{s}_{t+1} = \theta_t \Xi \quad (11)$$

Ξ has one column per output state dimension $[z_1, z_2, z_3, z_4, z_5, z_6, \tilde{y}]$ and one row per input feature across all $p + 1$ lag windows. Most entries are zero after thresholding.

Step 6. Island Search over (λ, p)

STLSQ requires two hyperparameters: threshold λ and lag order p . These are discovered by an island-model evolutionary search.

Island configuration

Parameter	Value
Number of islands K	4
Population per island	5 programs (best 5 retained)
Total iterations M	80
Migration interval G	every 10 iterations
Number of migration events	$M/G = 8$

Each island is seeded with a different bias to explore different regions of the search space:

Island	p range	Bias
0	[3, 4]	conservative, short lags
1	[5, 5]	default
2	[6, 8]	deep-lag exploration
3	[4, 5]	nonlinear, extra terms seeded

One iteration within an island

1. Sample parent (λ, p) from island population via tournament selection.
2. Mutate to produce child (λ', p') .
3. Run STLSQ with (λ', p') on training data.
4. Score child on validation loss: $f = -\text{MSE}_{\text{val}}$.
5. Add child to island; evict lowest-scoring program if population > 5 .

Migration (every $G = 10$ iterations)

Every 10 iterations, each island broadcasts its current best program to all other islands:

$$\forall i \neq j: \quad \text{Island}_j.\text{add}(\text{Island}_i.\text{best}()) \quad (12)$$

This is all-to-all migration. Migrated programs compete normally in the receiving island. The full search timeline:

$$\underbrace{[10 \text{ iters} \rightarrow \text{migrate}]}_{\times 8} \Rightarrow 80 \text{ total STLSQ evaluations per island, } 8 \text{ migration events.} \quad (13)$$

Final selection

After all 80 iterations, each island reports its best (λ^*, p^*) and corresponding Ξ^* . The winner is selected by validation score across all $K = 4$ islands:

$$(\lambda^*, p^*) = \arg \max_{k \in \{1, \dots, K\}} f_k^{\text{best}} \quad (14)$$

The winning sparse Ξ^* and its governing equation are passed to EpiHSD for evaluation on the held-out test set.

Step 7. EpiHSD Input

After the island search, the winning (λ^*, p^*) and the corresponding sparse Ξ^* define the governing equation. EpiHSD then takes the same latent state vector $\mathbf{S}$ and runs a three-level hierarchical sparse regression on it.

State vector entering EpiHSD

The state vector is unchanged from Step 3:

$$\mathbf{s}_t = [z_1, z_2, z_3, z_4, z_5, z_6, \tilde{y}]_t \in \mathbb{R}^7 \quad (15)$$

with $\tilde{y}_t = \log(1 + \max(0, y_t))$ and $L = 5$ lag steps.

Lag feature matrix for EpiHSD

EpiHSD concatenates $L + 1 = 6$ consecutive state blocks into one row (same construction as Step 4, but with a fixed $p = L = 5$):

$$\mathbf{x}_t = [\mathbf{s}_{t-L} \mid \cdots \mid \mathbf{s}_t] \in \mathbb{R}^{7 \times 6 = 42} \quad (16)$$

A polynomial feature map of degree 1 (linear, with bias) is then applied:

$$\boldsymbol{\theta}_t = \phi(\mathbf{x}_t) \in \mathbb{R}^{43} \quad (42 \text{ features} + 1 \text{ bias term}) \quad (17)$$

Stacking all valid windows across all provinces:

$$\boldsymbol{\Theta} \in \mathbb{R}^{T \times 43}, \quad \mathbf{Y} \in \mathbb{R}^{T \times 7}, \quad T \approx 8,000 \text{ (train years)} \quad (18)$$

Three-level hierarchical regression

EpiHSD fits STLSQ in three nested residual levels. Let $\mathbf{W} \in \mathbb{R}^{43 \times 7}$ denote a coefficient matrix at any level.

Level 1 — Global. Fit one equation on all provinces simultaneously:

$$\mathbf{Y} \approx \boldsymbol{\Theta} \mathbf{W}_{\text{global}} \quad (19)$$

Level 2 — Cluster. For each cluster c , compute residuals and fit a sparse correction:

$$\mathbf{R}^{(c)} = \mathbf{Y}^{(c)} - \boldsymbol{\Theta}^{(c)} \mathbf{W}_{\text{global}}, \quad \mathbf{R}^{(c)} \approx \boldsymbol{\Theta}^{(c)} \Delta \mathbf{W}^{(c)} \quad (20)$$

$$\mathbf{W}^{(c)} = \mathbf{W}_{\text{global}} + \Delta \mathbf{W}^{(c)} \quad (21)$$

Level 3 — Province. For each province p in cluster c , fit a further sparse correction on top of the cluster equation:

$$\mathbf{R}^{(p)} = \mathbf{Y}^{(p)} - \boldsymbol{\Theta}^{(p)} \mathbf{W}^{(c)}, \quad \mathbf{R}^{(p)} \approx \boldsymbol{\Theta}^{(p)} \Delta \mathbf{W}^{(p)} \quad (22)$$

$$\mathbf{W}^{(p)} = \mathbf{W}^{(c)} + \Delta \mathbf{W}^{(p)} \quad (23)$$

The province-level equation $\mathbf{W}^{(p)}$ is the final governing equation for province p .

Forecasting

Given the last $L + 1$ weeks of observed state, one-step prediction for province p is:

$$\hat{\mathbf{s}}_{t+1} = \phi(\mathbf{x}_t) \mathbf{W}^{(p)} \in \mathbb{R}^7 \quad (24)$$

The 7th entry $\hat{s}_{t+1,7}$ is the predicted log-incidence. Reporting converts back to raw scale: $\hat{y}_{t+1} = e^{\hat{s}_{t+1,7}} - 1$.

STLSQ thresholds at each level

Level	Latent dims $z_1 \dots z_6$	Incidence $\tilde{y}$
Global	$\lambda = 0.05$	$\lambda = 0.04$
Cluster	$\lambda = 0.05$	$\lambda = 0.04$
Province	$\lambda = 0.08$	$\lambda = 100$ (disabled)

The incidence threshold at province level is set to 100 (effectively infinity) so that province-level corrections refine only the latent dynamics $z_1 \dots z_6$, leaving incidence prediction entirely to the cluster equation.

Step 8. Final Output: Discovered Governing Equations

The full pipeline produces three nested coefficient matrices, one per hierarchical level. Each is a sparse $\mathbf{W} \in \mathbb{R}^{43 \times 7}$ whose surviving entries are the discovered equation of motion.

How to read an equation

Every surviving term has the form:

$$\alpha \cdot z_{k,t-\ell} \quad \text{or} \quad \alpha \cdot \tilde{y}_{t-\ell} \quad (25)$$

where α is the coefficient, $k \in \{1, \dots, 6\}$ is the latent dimension, and $\ell \in \{0, 1, \dots, 5\}$ is the lag. The output is one of the 7 state dimensions at $t + 1$.

Level 1: Global equation (example — incidence column)

The globally discovered incidence equation is:

$$\tilde{y}_{t+1} = 0.436 \tilde{y}_t + 0.205 \tilde{y}_{t-1} + 0.132 \tilde{y}_{t-2} + 0.060 \tilde{y}_{t-3} + 0.052 \tilde{y}_{t-5} - 0.053 z_{4,t} - 0.050 z_{6,t} \quad (26)$$

This says that across all 32 provinces, next-week log-incidence is driven primarily by its own autoregressive history (lags 0-3 and 5), with small corrections from latent dimensions z_4 and z_6 at the current week.

Level 2: Cluster delta equations

Each cluster c adds a sparse correction $\Delta \mathbf{W}^{(c)}$ on top of the global equation. For example, Cluster 3 (11 provinces) has a minimal incidence correction:

$$\Delta \tilde{y}_{t+1}^{(3)} = 0.055 \tilde{y}_t + 0.060 \tilde{y}_{t-1} - 0.116 \tilde{y}_{t-2} + 0.050 z_{4,t-3} + 0.045 z_{1,t-1} + 0.062 z_{4,t-1} \quad (27)$$

Cluster 4 (the best-performing cluster, test $R^2 = 0.53$) has an even sparser correction, relying almost entirely on the global autoregressive structure:

$$\Delta \tilde{y}_{t+1}^{(4)} = 0.067 \tilde{y}_t - 0.080 \tilde{y}_{t-3} + 0.115 \tilde{y}_{t-4} - 0.077 \tilde{y}_{t-5} \quad (28)$$

The sparsity of cluster corrections is by design. Clusters with dense $\Delta \mathbf{W}^{(c)}$ (e.g. Cluster 0 with 103 active delta terms) indicate that the global equation is a poor fit for those provinces and more regional structure is needed.

Level 3: Province equation

The final province-level equation is the sum of all three levels:

$$\mathbf{W}^{(p)} = \mathbf{W}_{\text{global}} + \Delta \mathbf{W}^{(c(p))} + \Delta \mathbf{W}^{(p)} \quad (29)$$

Province-level corrections refine only the latent dimensions $z_1 \dots z_6$. Incidence prediction is frozen at the cluster level ($\lambda_{\text{inc}}^{(p)} = 100$).

Performance summary

Method	Val R^2 (inc)	Test R^2 (inc)	Interpretable	Hierarchical
Linear Regression	—	0.611	partial	no
Lasso	—	0.689	partial	no
Random Forest	—	0.666	no	no
Gradient Boosting	—	0.665	no	no
GRU (neural)	0.160	0.290	no	no
EpiHSD baseline	0.230	0.397	yes	yes
EpiHSD + FunSearch	0.496	0.394	yes	yes
EpiHSD + FS cluster 4	0.691	0.531	yes	yes

The simple ML baselines achieve higher raw incidence R^2 by fitting directly to incidence history without any physical structure. EpiHSD operates in a 6-dimensional latent space that encodes climate, spatial, and epidemiological dynamics jointly. Its advantage is not peak R^2 on a single split but the interpretability of the discovered equations, the hierarchical decomposition of global versus regional dynamics, and the generalisable structure of the latent state.